

vLLM Audit Implementation - Deployment Quick Reference

🚀 Ready to Deploy

All implementation work is **COMPLETE**. This is your deployment checklist.

⚡ Quick Deploy (2 Steps)

Step 1: Restart API

```
docker compose restart api
```

Step 2: Rebuild UI

```
cd src/ui_launchers/Karen-AI-Theme  
npm run build  
docker compose restart web
```

Done! The fixes are now live.

✓ Verify Deployment

Test 1: Check API Health

```
curl http://localhost:8000/api/health/providers/all | jq
```

Expected: List of all providers with health status

Test 2: Test vLLM Diagnostics

```
curl http://localhost:8000/api/health/providers/vllm | jq
```

Expected: vLLM health details including connectivity and model tests

Test 3: Test Fallback (Disable Gemini)

```
# 1. Comment out GEMINI_API_KEY in .env
# 2. Restart API: docker compose restart api
# 3. Make request:
curl -sS http://localhost:8000/api/chat/chat \
  -H "Content-Type: application/json" \
  -d '{"message":"Hello","provider":"gemini"}' | jq '.metadata'
```

Expected Metadata:

```
{
  "requested_provider": "gemini",
  "actual_provider": "builtin_vllm",
  "runtime_engine": "vllm",
  "response_source": "live_model",
  "fallback_level": 1,
  "degraded_mode": true
}
```

Test 4: Check Logs

```
docker compose logs api --tail=50 | grep -i "fallback"
```

Expected: Log entries showing fallback attempts and success

📊 Monitor After Deployment

Prometheus Metrics

PROF

```
curl http://localhost:8000/metrics | grep
kari_llm_provider_fallbacks_total
```

Key Metrics to Watch

- `kari_llm_provider_fallbacks_total` - Fallback count (should be low)
 - `kari_llm_provider_failures_total` - Provider failures
 - `kari_llm_provider_latency_seconds` - Response times
-

🔧 Troubleshooting

Issue: Fallback Not Working

Check:

1. API restarted? `docker compose restart api`
2. Logs show fallback attempt? `docker compose logs api | grep fallback`
3. vLLM server running? `curl http://localhost:8001/health`

Solution: See [docs/VLLM_TROUBLESHOOTING.md](#)

Issue: Ollama Still Shows "Legacy Runtime"

Check:

1. UI rebuilt? `cd src/ui_launchers/Karen-AI-Theme && npm run build`
2. Web service restarted? `docker compose restart web`
3. Browser cache cleared? Hard refresh (Ctrl+Shift+R)

Solution: Rebuild UI and clear browser cache

Issue: Wrong Provider in Metadata

Check:

1. API restarted with new code? `docker compose restart api`
2. Request includes provider? `{"provider":"gemini"}`
3. Logs show provider selection? `docker compose logs api | grep "provider selection"`

Solution: Ensure API has latest code



Documentation

Quick Guides

- **Quick Start:** [docs/VLLM_AUDIT_QUICKSTART.md](#)
- **Testing:** [docs/VLLM_TESTING_GUIDE.md](#)
- **Deployment:** [docs/VLLM_DEPLOYMENT_GUIDE.md](#)
- **Troubleshooting:** [docs/VLLM_TROUBLESHOOTING.md](#)

Complete Documentation

- **Full Implementation:** [VLLM_AUDIT_IMPLEMENTATION_COMPLETE.md](#)
- **Phase 4 Enhancements:** [docs/PHASE4_ENHANCEMENTS.md](#)
- **Audit Specification:** [docs/VLLM_RUNTIME_AUDIT.md](#)



What Changed

Backend (5 Files)

1. [response_synth.py](#) - Calls fallback method on provider failure

2. `orchestration_state.py` - Added `llm_metadata` field
3. `runtime.py` - Extracts and returns fallback metadata
4. `vllm_runtime.py` - Added fallback logging
5. `health/providers.py` - **NEW** - Diagnostics endpoints

Frontend (1 File)

1. `chat-response.ts` - Removed Ollama from legacy aliases

Tests (2 Files)

1. `test_fallback_chain.py` - **NEW** - 8 integration tests
2. `test_vllm_smoke.py` - **NEW** - vLLM smoke tests

🔧 Optional Enhancements

Recommended

- **UI Fallback Indicator** - Visual feedback when fallback occurs
 - See `docs/PHASE4_ENHANCEMENTS.md` for implementation

Nice to Have

- **Environment Variable Config** - Tune circuit breaker via `.env`
- **Grafana Dashboard** - Visual monitoring
- **YAML Config File** - Centralized configuration

Note: All core functionality works without these enhancements.

☆☆ What Works Now

✓ Automatic Fallback

When Gemini fails → tries vLLM → tries Transformers → emergency response

✓ Accurate Metadata

Every response shows which provider actually generated it

✓ Comprehensive Logging

Every fallback attempt is logged with reason and outcome

✓ Health Monitoring

Diagnostics endpoints test provider connectivity and generation

✓ Resilience

Support

Common Commands

```
# Restart everything
docker compose restart

# Check API logs
docker compose logs api --tail=100

# Check web logs
docker compose logs web --tail=100

# Test vLLM server
curl http://localhost:8001/health

# Test Karen API
curl http://localhost:8000/api/health
```

Documentation Index

- All guides in [docs/](#) directory
- Implementation report: [VLLM_AUDIT_IMPLEMENTATION_COMPLETE.md](#)
- This quick reference: [DEPLOYMENT_QUICK_REFERENCE.md](#)

🎯 Success Criteria

Deployment is successful when:

- ✓ API restarts without errors
- ✓ UI builds and serves correctly
- ✓ [/api/health/providers/all](#) returns provider list
- ✓ Fallback test shows [actual_provider](#) different from [requested_provider](#)
- ✓ Logs show fallback attempts
- ✓ Ollama no longer shows "legacy runtime" error

Last Updated: 2026-04-27

Status: ✓ READY TO DEPLOY

Estimated Deploy Time: 5 minutes

Risk Level: Low (backward compatible)

For detailed information, see [VLLM_AUDIT_IMPLEMENTATION_COMPLETE.md](#)